

Trader Performance Analysis Based on Bitcoin Market Sentiment

Executive Summary

This project analyzes the relationship between Bitcoin market sentiment and trader performance using the Bitcoin Fear & Greed Index and Hyperliquid historical trading data. The objective was to investigate whether market sentiment influences trader profitability, trading activity, position sizing, and overall trading behavior.

The analysis combined 211,224 Hyperliquid trading records with 2,644 daily sentiment observations from the Fear & Greed Index. Results reveal that market sentiment has a measurable impact on trading outcomes. Extreme Greed periods generated the highest average profitability and win rates, while Fear periods produced the highest trading activity, largest position sizes, and highest transaction costs.

A particularly important finding is that successful traders do not necessarily maintain high win rates. Instead, profitability appears to be driven by effective risk management and favorable risk-reward ratios.

1. Introduction

Financial markets are heavily influenced by investor psychology and market sentiment. The Bitcoin Fear & Greed Index measures overall market sentiment by classifying market conditions into categories such as Extreme Fear, Fear, Neutral, Greed, and Extreme Greed.

The objective of this study is to determine how trader performance varies across different market sentiment conditions and to identify behavioral patterns associated with successful trading.

Objectives

- Analyze trader profitability under different sentiment conditions.
 - Evaluate the relationship between sentiment and win rate.
 - Examine trading activity and position sizing behavior.
 - Identify characteristics of top-performing traders.
 - Provide actionable insights for traders and trading platforms.
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2. Dataset Description

2.1 Bitcoin Fear & Greed Index

Total Records: 2,644

Columns:

Column	Description
timestamp	Unix timestamp
value	Fear & Greed score
classification	Market sentiment category
date	Observation date

Sentiment Categories:

- Extreme Fear
- Fear
- Neutral
- Greed
- Extreme Greed

2.2 Hyperliquid Historical Trading Data

Total Records: 211,224 Trades

Columns Used:

Column	Description
Account	Trader identifier
Coin	Cryptocurrency traded
Size USD	Trade value in USD
Closed PnL	Realized profit/loss
Fee	Trading fee
Direction	Trade direction
Timestamp IST	Trade timestamp

No missing values were found in either dataset.

3. Data Preparation

The following preprocessing steps were performed:

Step 1: Date Standardization

- Converted sentiment dataset dates into datetime format.
- Converted Hyperliquid timestamps into datetime format.
- Extracted trading dates from timestamps.

Step 2: Dataset Merging

The two datasets were merged using the date field.

Result:

- Every trade was assigned a corresponding market sentiment label.

Step 3: Feature Engineering

A binary win indicator was created:

Win = 1 if Closed PnL > 0

Win = 0 otherwise

This enabled sentiment-wise win rate calculations.

4. Exploratory Data Analysis

4.1 Sentiment Distribution

Trading activity was distributed across five sentiment categories:

Sentiment	Trade Count
Fear	61,837
Greed	50,303
Extreme Greed	39,992

Sentiment	Trade Count
Neutral	37,686
Extreme Fear	21,400

Observation

Fear periods generated the highest trading activity, accounting for the largest number of trades.

This suggests traders become more active during uncertain market conditions.

4.2 Profitability Analysis

Average Closed PnL by sentiment:

Sentiment	Average PnL
Extreme Greed	67.89
Fear	54.29
Greed	42.74
Extreme Fear	34.54
Neutral	34.31

Observation

Extreme Greed generated the highest average profit per trade.

Interpretation

Bullish market environments appear to create more favorable trading opportunities and allow traders to capture larger gains.

4.3 Win Rate Analysis

Sentiment	Win Rate
Extreme Greed	46.49%
Fear	42.08%
Neutral	39.70%

Sentiment	Win Rate
Greed	38.48%
Extreme Fear	37.06%

Observation

The highest win rate occurs during Extreme Greed periods.

Interpretation

Market optimism appears to improve trading outcomes and increase the probability of successful trades.

4.4 Position Size Analysis

Average position size:

Sentiment	Avg Position Size (USD)
Fear	7,816
Greed	5,737
Extreme Fear	5,350
Neutral	4,783
Extreme Greed	3,112

Observation

Traders deployed the largest positions during Fear periods.

Interpretation

This result suggests traders may be aggressively buying perceived discounts during market downturns.

Interestingly, the largest position sizes did not correspond to the highest profitability.

4.5 Fee Analysis

Average trading fee:

Sentiment	Avg Fee
Fear	1.495
Greed	1.254
Extreme Fear	1.116
Neutral	1.045
Extreme Greed	0.676

Observation

Fear periods generated the highest transaction costs.

Interpretation

Higher trading activity and larger positions during Fear contribute to increased fees and potentially lower trading efficiency.

5. Top Trader Analysis

The most profitable trader generated:

- Total Profit: \$2.14 Million
- Win Rate: 33.7%

Another trader achieved:

- Win Rate: 81.1%
- Total Profit: \$379 Thousand

Key Insight

A high win rate does not necessarily result in higher profitability.

This indicates that successful traders focus on:

- Risk management
- Position sizing
- Capturing large profitable trades

rather than maximizing win frequency.

6. Key Findings

Finding 1

Extreme Greed periods generate the highest profitability and win rates.

Finding 2

Fear periods attract the highest trading activity.

Finding 3

Traders take significantly larger positions during Fear periods.

Finding 4

Fear periods incur the highest transaction costs.

Finding 5

Large position sizes do not guarantee superior performance.

Finding 6

Risk-reward management is more important than win rate for long-term profitability.

7. Recommendations

For Traders

- Avoid excessive position sizing during Fear periods.
- Focus on maintaining favorable risk-reward ratios.
- Avoid overtrading during volatile market conditions.
- Prioritize trade quality over trade quantity.

For Trading Platforms

- Implement risk alerts during high-fear periods.
 - Provide analytics focused on trading efficiency.
 - Educate users on risk-reward optimization.
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8. Conclusion

This study demonstrates a strong relationship between Bitcoin market sentiment and trader performance.

Extreme Greed periods were associated with the highest profitability and win rates, while Fear periods generated the highest trading activity, position sizes, and transaction costs.

The analysis further revealed that trader success is not determined by win rate alone. Instead, profitability depends largely on effective risk management and the ability to maximize gains from winning trades.

These findings provide valuable insights for traders seeking to optimize performance across different market conditions and for platforms looking to enhance trader decision-making through data-driven tools.